

[BUG] Hermitian operator fails during `expval` with TensorFlow

<https://github.com/PennyLaneAI/pennylane/issues/2892>

ROW 107

[BUG] Hermitian operator fails during `expval` with TensorFlow #2892

New issue

Closed

#2895



jackaraz opened on Aug 4, 2022

Expected behavior

Hi, I'm trying to compute the expectation value of a Hermitian operator which works with the NumPy backend:

```
dev = qml.device("default.qubit", wires=2)

@qml.qnode(dev)
def circuit(K):
    return qml.expval(qml.Hermitian(K, wires=range(2)))
state = tf.constant([[1,2,3,4]], dtype=tf.float64)
K = tf.transpose(state)@state / 16.
circuit(K.numpy())
# tensor(0.0625, requires_grad=True)
```

However, when initiated with TensorFlow backend, the same code raises an `AttributeError` (see the details below). I would like to be able to take the gradient of this operator eventually. However, I'm afraid it won't be possible with TensorFlow if I convert it to NumPy to compute the expectation value. As far as I know, it won't compute the gradient if I take the state and compute the expectation value outside the circuit since there will be complex numbers. Would it work correctly if I only used the real portion of the expectation value after computing it out of the circuit? Any workaround would be appreciated if there is no quick fix.

Thanks!

Actual behavior

Code crashes and raises an `AttributeError`.

Additional information

No response

Source code

```
import pennylane as qml
import tensorflow as tf

dev = qml.device("default.qubit.tf", wires=2)

@qml.qnode(dev, interface="tf", diff_method="backprop")
def circuit(K):
    return qml.expval(qml.Hermitian(K, wires=range(2)))
state = tf.constant([[1,2,3,4]], dtype=tf.float64)
K = tf.transpose(state)@state / 16.
circuit(K)

### Second option

dev2 = qml.device("default.qubit.tf", wires=2)

@qml.qnode(dev2, interface="tf", diff_method="backprop")
def circuit2(op):
    return qml.expval(op)
circuit2(qml.Hermitian(K, wires=range(2)))
```

Tracebacks

```
AttributeError                                Traceback (most recent call last)
Input In [46], in <cell line: 3>()
      1 state = tf.constant([[1,2,3,4]], dtype=tf.float64)
      2 K = tf.transpose(state)@state / 16.
----> 3 circuit(K)

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/qnode.py:619, in QNode.__call__(self, *args,
      612 using_custom_cache = (
      613     hasattr(cache, "__getitem__")
```

Assignees

No one assigned

Labels

bug 🐛

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

🔗 Update the Hermiticity check of
`qml.Hermitian` to use `qml.math`
PennyLaneAI/pennylane

Notifications

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Participants



Zo

```

613         hasattr(cache, "__getitem__"),
614         and hasattr(cache, "__setitem__"),
615         and hasattr(cache, "__delitem__")
616     )
617 self._tape_cached = using_custom_cache and self.tape.hash in cache
--> 619 res = qml.execute(
620     [self.tape],
621     device=self.device,
622     gradient_fn=self.gradient_fn,
623     interface=self.interface,
624     gradient_kwargs=self.gradient_kwargs,
625     override_shots=override_shots,
626     **self.execute_kwargs,
627 )
629 if isinstance(res, (tuple, list)) and len(res) == 1:
630     # If a device batch transform was applied, we need to 'unpack'
631     # the returned tuple/list to a float.
632     (...)
638     # TODO: find a more explicit way of determining that a batch transform
639     # was applied.
641     res = res[0]

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/interfaces/execution.py:344, in execute(tapes)
340     return batch_fn(res)
342 if gradient_fn == "backprop" or interface is None:
343     return batch_fn(
--> 344         qml.interfaces.cache_execute(
345             batch_execute, cache, return_tuple=False, expand_fn=expand_fn
346         )(tapes)
347     )
349 # the default execution function is batch_execute
350 execute_fn = qml.interfaces.cache_execute(batch_execute, cache, expand_fn=expand_fn)

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/interfaces/execution.py:172, in cache_execute(tapes)
168     return (res, []) if return_tuple else res
170 else:
171     # execute all unique tapes that do not exist in the cache
--> 172     res = fn(execution_tapes.values(), **kwargs)
174 final_res = []
176 for i, tape in enumerate(tapes):

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/interfaces/execution.py:97, in cache_execute.fn(tapes, **kwargs)
95 def fn(tapes, **kwargs): # pylint: disable=function-redefined
96     tapes = [expand_fn(tape) for tape in tapes]
--> 97     return original_fn(tapes, **kwargs)

File ~/packages/miniforge3/lib/python3.9/contextlib.py:79, in ContextDecorator.__call___.<locals>.inner(*args, **kwargs)
76 @wraps(func)
77 def inner(*args, **kwargs):
78     with self._recreate_cm():
--> 79         return func(*args, **kwargs)

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/_qubit_device.py:355, in QubitDevice.batch_execute(circuits)
350 for circuit in circuits:
351     # we need to reset the device here, else it will
352     # not start the next computation in the zero state
353     self.reset()
--> 355     res = self.execute(circuit)
356     results.append(res)
358 if self.tracker.active:

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/_qubit_device.py:257, in QubitDevice.execute(circuit)
254 self.check_validity(circuit.operations, circuit.observables)
256 # apply all circuit operations
--> 257 self.apply(circuit.operations, rotations=circuit.diagonalizing_gates, **kwargs)
259 # generate computational basis samples
260 if self.shots is not None or circuit.is_sampled:

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/tape/tape.py:1319, in QuantumTape.diagonalizing_gates(observable)
1315 for observable in self.observables:
1316     # some observables do not have diagonalizing gates,
1317     # in which case we just don't append any
1318     try:
--> 1319         rotation_gates.extend(observable.diagonalizing_gates())
1320     except qml.operation.DiagGatesUndefinedError:
1321         pass

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/ops/qubit/observables.py:171, in Hermitian.diagonalizing_gates(observable)
164 """Return the gate set that diagonalizes a circuit according to the
165 specified Hermitian observable.
166
167 Returns:
168     list: list containing the gates diagonalizing the Hermitian observable
169 """
170 # note: compute_diagonalizing_gates has a custom signature, which is why we overwrite this method

```

```

--> 171 return self.compute_diagonalizing_gates(self.eigendecomposition["eigvec"], self.wires)

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/ops/qubit/observables.py:113, in Hermitian.eigendecomposition
 101 @property
 102 def eigendecomposition(self):
 103     """Return the eigendecomposition of the matrix specified by the Hermitian observable.
 104
 105     This method uses pre-stored eigenvalues for standard observables where
 106     (...)
 111     dict[str, array]: dictionary containing the eigenvalues and the eigenvectors of the Hermitian
 112     """
--> 113 Hmat = self.matrix()
 114 Hmat = qml.math.to_numpy(Hmat)
 115 Hkey = tuple(Hmat.flatten().tolist())

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/operation.py:585, in Operator.matrix(self, wire_order)
 565 def matrix(self, wire_order=None):
 566     r"""Representation of the operator as a matrix in the computational basis.
 567
 568     If ``wire_order`` is provided, the numerical representation considers the position of the
 569     (...)
 583     tensor_like: matrix representation
 584     """
--> 585 canonical_matrix = self.compute_matrix(*self.parameters, **self.hyperparameters)
 587 if wire_order is None or self.wires == Wires(wire_order):
 588     return canonical_matrix

File ~/packages/miniforge3/lib/python3.9/site-packages/pennylane/ops/qubit/observables.py:96, in Hermitian.conj
 93 if A.shape[0] != A.shape[1]:
 94     raise ValueError("Observable must be a square matrix.")
--> 96 if not qml.math.allclose(A, A.conj().T):
 97     raise ValueError("Observable must be Hermitian.")
 99 return A

File ~/packages/miniforge3/lib/python3.9/site-packages/tensorflow/python/framework/ops.py:513, in Tensor.__getattr__
 505 if name in {"T", "astype", "ravel", "transpose", "reshape", "clip", "size",
 506            "tolist", "data"}:
 507     # TODO(wangpeng): Export the enable_numpy_behavior knob
 508     raise AttributeError("""
 509     '{}' object has no attribute '{}'.
 510     If you are looking for numpy-related methods, please run the following:
 511     from tensorflow.python.ops.numpy_ops import np_config
 512     np_config.enable_numpy_behavior()""".format(type(self).__name__, name))
--> 513 self.__getattr__(name)

AttributeError: 'tensorflow.python.framework.ops.EagerTensor' object has no attribute 'conj'

```

System information

```

Name: PennyLane
Version: 0.24.0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /Users/jackaraz/packages/miniforge3/lib/python3.9/site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, retworkx, scipy, semar
Required-by: PennyLane-Lightning, qhbm

Platform info:      macOS-12.5-arm64-arm-64bit
Python version:     3.9.10
Numpy version:      1.21.6
Scipy version:      1.8.0
Installed devices:
- lightning.qubit (PennyLane-Lightning-0.24.0)
- default.gaussian (PennyLane-0.24.0)
- default.mixed (PennyLane-0.24.0)
- default.qubit (PennyLane-0.24.0)
- default.qubit.autograd (PennyLane-0.24.0)
- default.qubit.jax (PennyLane-0.24.0)
- default.qubit.tf (PennyLane-0.24.0)
- default.qubit.torch (PennyLane-0.24.0)

```

Existing GitHub issues

```
default.qubit (PennyLane-0.24.0)
- default.qubit.autograd (PennyLane-0.24.0)
- default.qubit.jax (PennyLane-0.24.0)
- default.qubit.tf (PennyLane-0.24.0)
- default.qubit.torch (PennyLane-0.24.0)
```

Existing GitHub issues

- [X] I have searched existing GitHub issues to make sure the issue does not already exist.



jackaraz added **bug** on Aug 4, 2022



antalszava on Aug 5, 2022

Contributor ...

Hi [@jackaraz](#), thanks for reporting this! We'll be looking into this. :)



antalszava mentioned this on Aug 5, 2022

Update the Hermiticity check of `qml.Hermitian` to use `qml.math` #2895



AlbertMitjans on Aug 16, 2022

Contributor ...

I believe this is solved. Closing issue.



AlbertMitjans closed this as **completed** on Aug 16, 2022

timmysilv mentioned this on Feb 13, 2023

[BUG] Update validation check in `qml.Hermitian` to support multiple interfaces #2132

Update the Hermiticity check of `qml.Hermitian` to use `qml.math` #2895

<> Code

Merged antalszava merged 6 commits into `v0.25.0-rc0` from `fix_2892` on Aug 8, 2022

Conversation 6 Commits 6 Checks 0 Files changed 3 +26 -3



antalszava commented on Aug 5, 2022

Contributor

Reworks the Hermiticity check in `qml.Hermitian` by using `qml.math` because calling `.conj()` on an `EagerTensor` from TF raises an error.

Closes #2892



fix

b26d074



github-actions bot commented on Aug 5, 2022

Contributor

Hello. You may have forgotten to update the changelog!
Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



antalszava added 2 commits 3 years ago

changelog

ea5c1c1

lint

de717ea

antalszava requested a review from Jaybsoni 3 years ago



antalszava commented on Aug 5, 2022

View reviewed changes

pennylane/ops/qubit/observables.py

Show resolved



Jaybsoni approved these changes on Aug 5, 2022

View reviewed changes

Jaybsoni left a comment

Contributor

Looks good to me! I am surprised that tf can't natively resolve this, but autoray can!



Jaybsoni commented on Aug 5, 2022

Contributor

Maybe add a more descriptive title to the PR?



Reviewers

Jaybsoni



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[BUG] Hermitian operator fails during expva...

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2 participants

